

Lecture 06: Actuators/Solenoids



Overview

- **Actuators / Solenoids Modeling**
- **Translational Motion**
- **Magnetic Field Diffusion**
- **Workshop 06: Actuator Example (with Parametric analysis)**
- **Appendix 6.1 – Material Definitions**
- **Appendix 6.2 - Losses**

Actuators / Solenoids Modeling

- **Actuators / Solenoids**
 - Most often the actuators/solenoids geometries are axial-cylindrical, thus it is enough to model the devices in Maxwell through a 2D Cylindrical around Z Design.
 - This fact presents some major advantages:
 - Model is easier to be created
 - It saves simulation time
 - One can get better accuracy on equal simulation time with respect to a full 3D model
 - For more complicated geometries a complete 3D Design may be needed
 - It is possible to create the model directly in Maxwell or
 - It is possible to import the geometry from external CAD tools

Common Modeling approached with FEA

- **Static Models:**

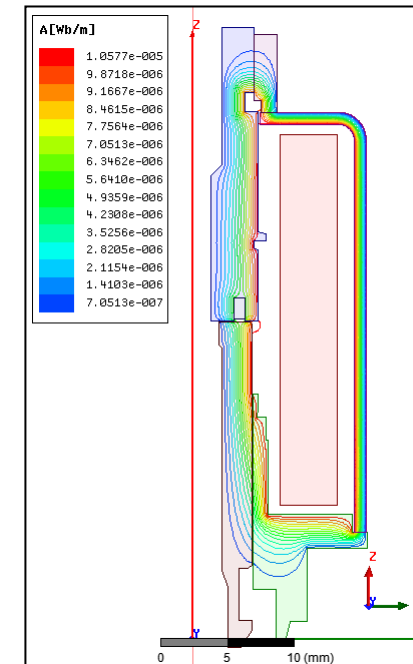
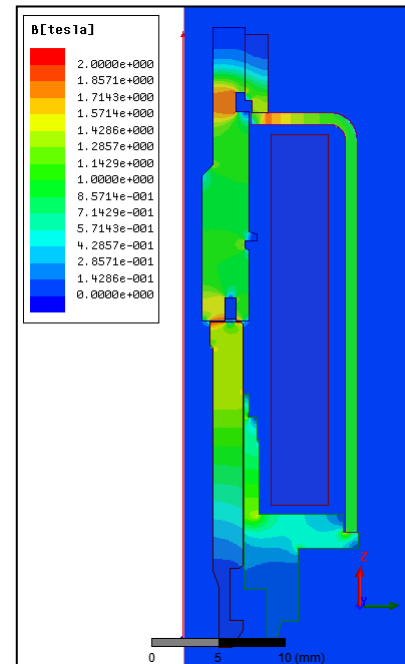
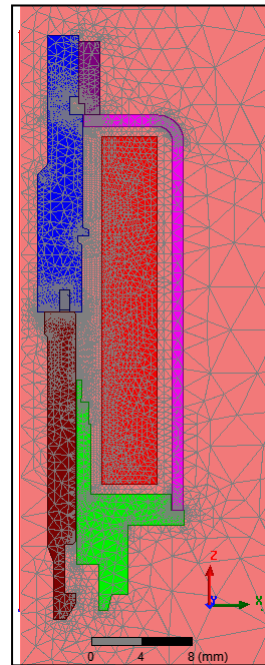
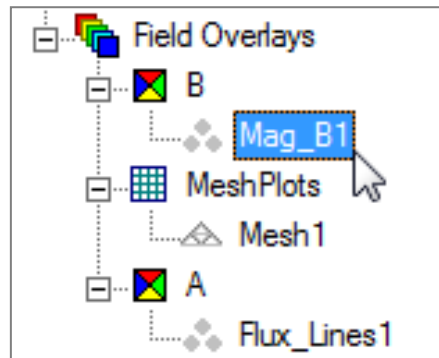
- Using the *Magnetostatic Solver*, 2D or 3D FEA provides users the ability to create force vs stroke vs current curves, sometimes called “pull curves,” and useful for simulation studies such as pole shaping.
- This data is often used to correlate static force lab measurements,
- No eddy effects or magnetic diffusion considered
- Note: Measurement hysteresis may exist due to friction in the test fixture.

- **Transient Models:**

- Using the *Transient Solver*, 2D or 3D FEA provides users the ability to model current, force, position vs time.
- This approach solves both the electrical and mechanical transients (such as eddy effects and magnetic diffusion).
- Requires information about the drive circuit and control, as well as the mechanical load that is connected to the armature, and also useful for pole shaping and coil design.

Suggested Modeling Practice

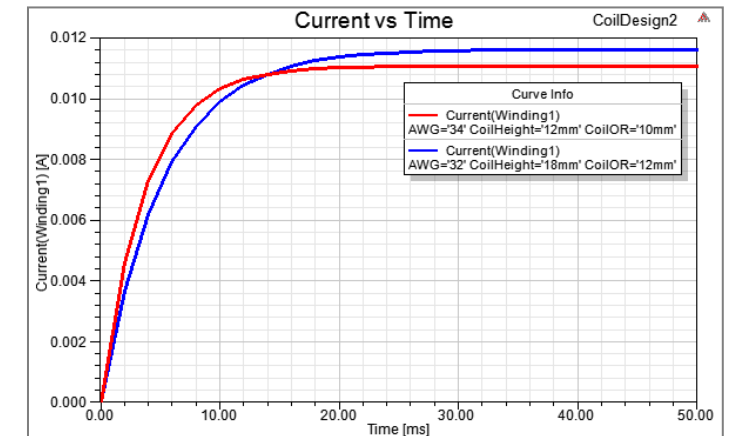
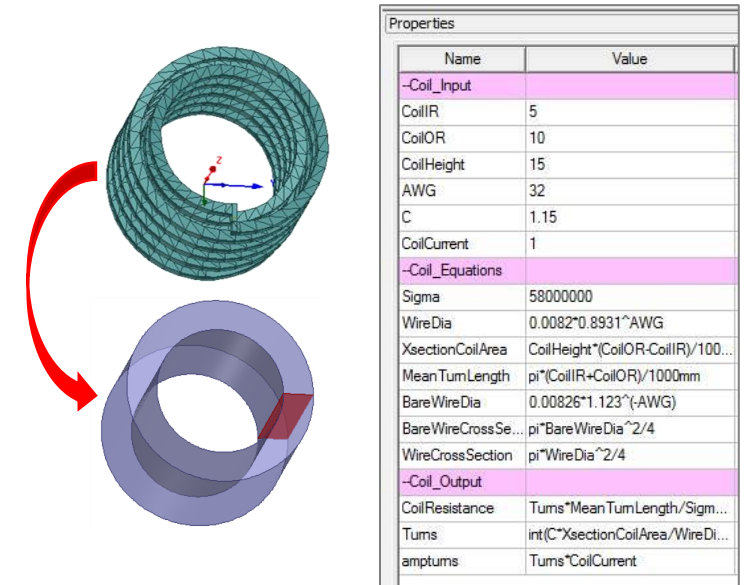
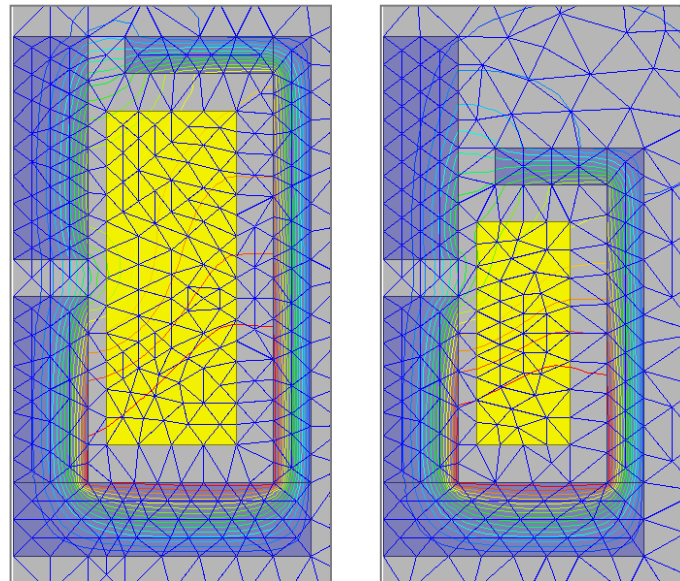
- It is always recommended that users create the following plots: **Mesh**, **Flux Density**, **Flux Lines**
- Animations of these plots versus modeled variables or time also provide design insight.
- These plots help ensure that the model is defined correctly by the user – for example:
 - Boundaries and excitation values and directions, materials, units, number of turns, etc.
 - Saturation occurs where expected, leakage and fringing is observed, mesh density is created where needed, etc.



Incorporating Coil Design and FEA

- Usually, multi-turn coils are modeled as equivalent ‘lumped’ objects in FEA. (In certain cases, the coils turns can be explicitly modeled)
- Coil design equations can be incorporated directly in the magnetic FEA model.
- For Parametric or Optimization studies, as the coil size changes, the coil performance is automatically calculated.
- Example:

Parameter	Design #1	Design #2
Coil Height (mm)	18	12
Coil Radius (mm)	12	10
AWG	32	34
Turns	2991	2238
Resistance (Ohms)	86.15	90.46
Inductance (mH)	372	255
Steady State Current (mA)	11.6	11.1
Time Constant (L/R)	.0026	.0016

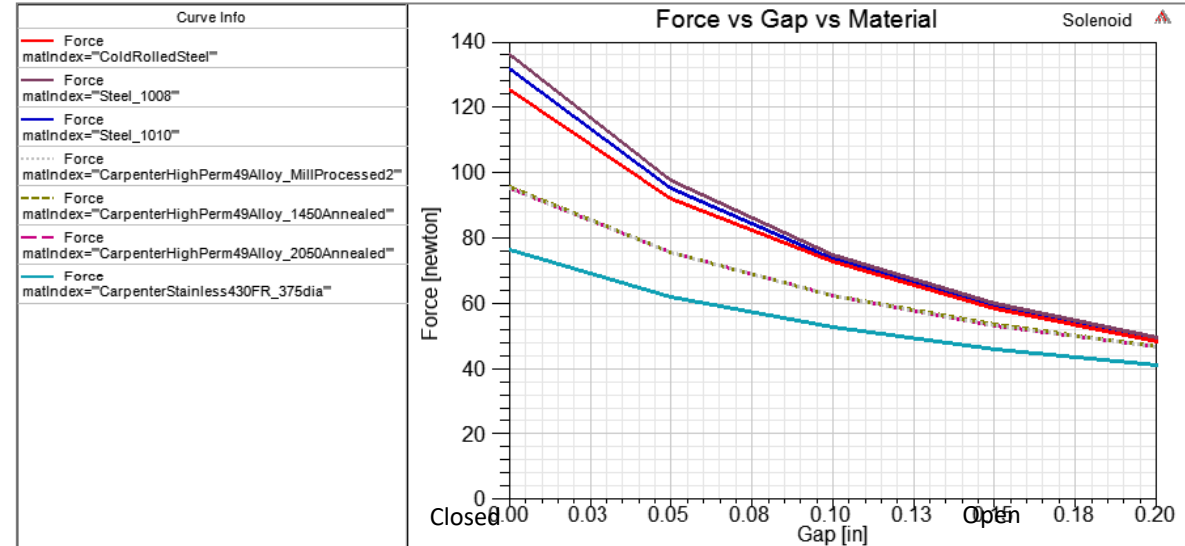
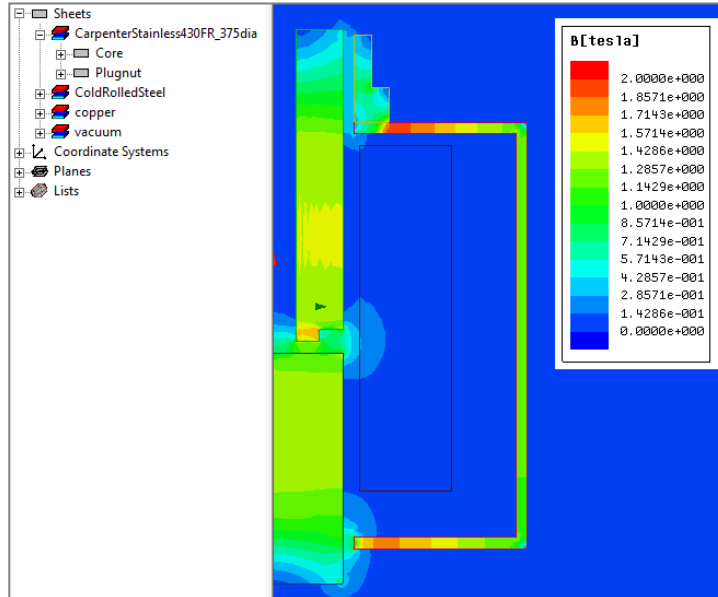
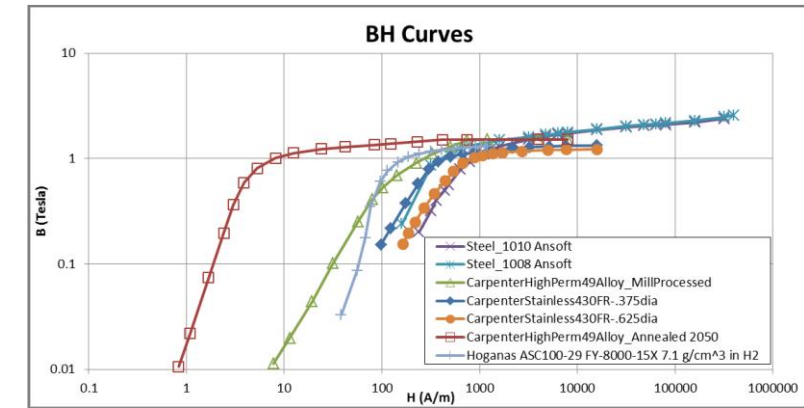


Comments on Modeling Magnetic Gaps

- In some cases, in the “armature closed” or “seal” position, a force calculation is required.
- Special considerations should be made when modeling a zero mechanical gap with FEA models.
 - How two pieces of ferrous material mate together can impact the measured force.
 - Surface finish, smoothness, flatness, dirt, etc. can create an effective magnetic gap between these two pieces.
 - Modeling ideal shapes in FEA with zero gap between them may not be realizable with actual parts.
- Thus, the concept of an “Equivalent Magnetic Gap” is often used.
 - A common practice is to use an offset position of the armature between values of 0.05 and 0.1 mm.
 - This value may vary depending on the above mentioned variables as well as the aspect ratio of the diameter to the gap, or correlation to measurement.

Material Study Using FEA

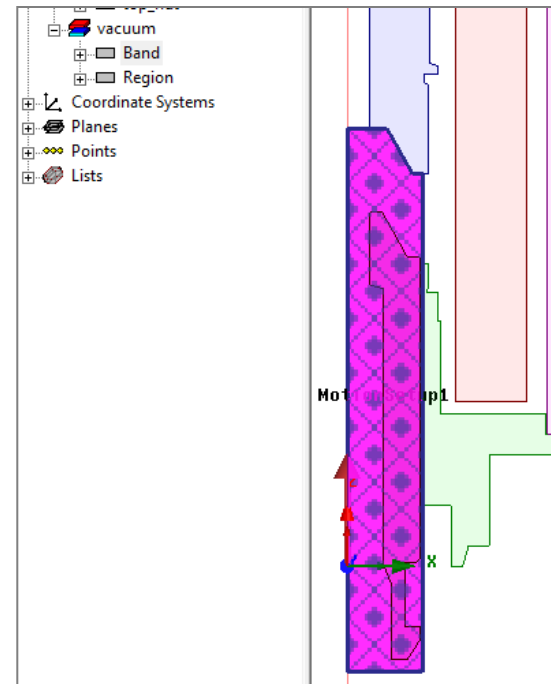
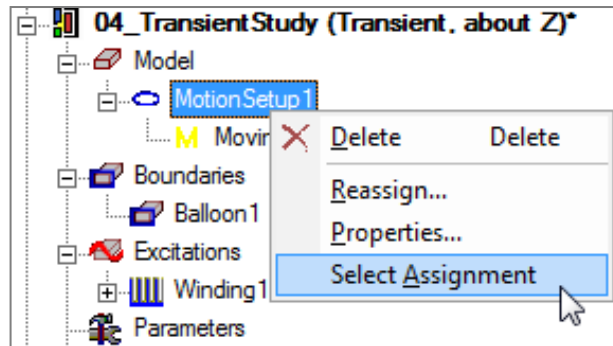
- Hard and soft materials can be defined with B vs H curves that can vary greatly by composition, heat treatments, machining, etc.
- Including different materials in a Parametric Sweep (using Array Index) allows users to quickly evaluate material performance and effect on Force or Inductance.



Translational Motion

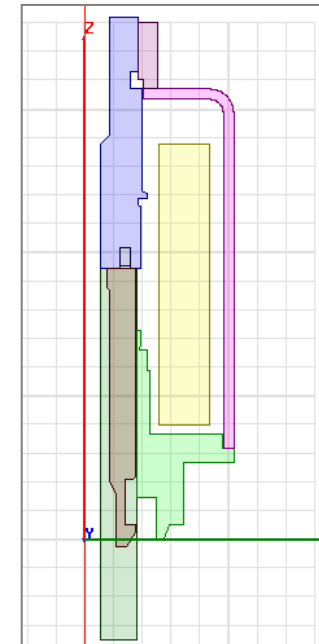
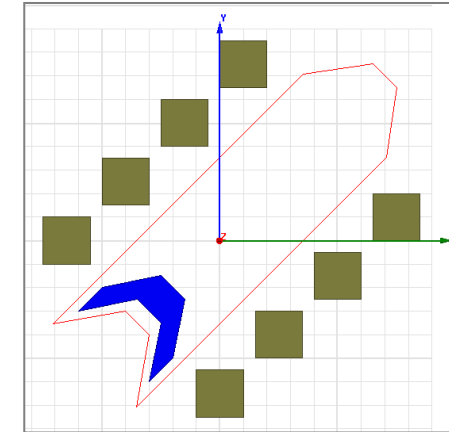
Translational Motion Best Practices – Band Objects

- The Motion Setup in Maxwell requires the definition of a so called “**Container**” Band object.
- This Band object encloses all the moving objects – when motion occurs, the area between the moving objects and the Band re-meshes at every time step, and all other mesh is held constant.



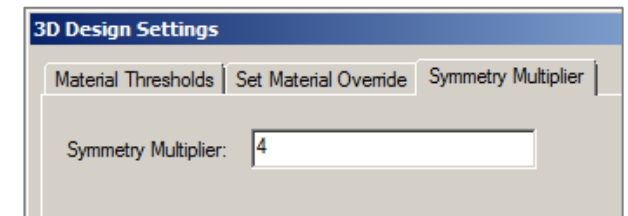
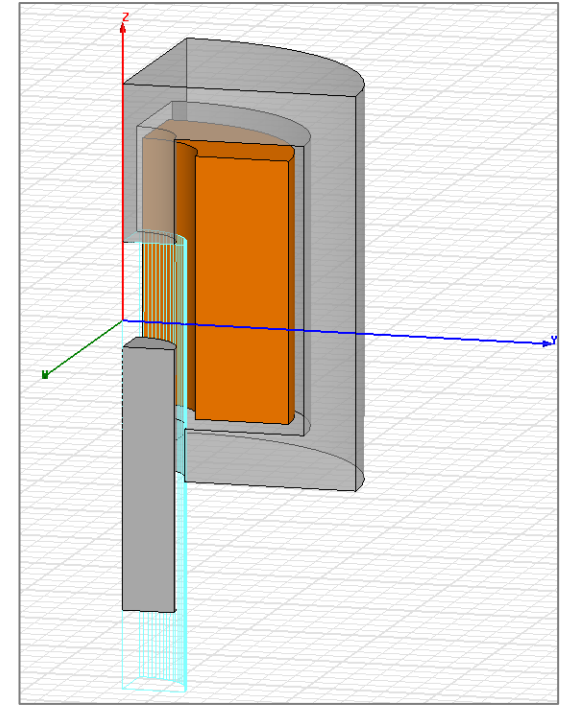
/ Translational Motion in 2D XY and RZ

- In **2D XY**, the motion can be along the X- or Y-axis for any defined coordinate system. The band and shape of the objects of motion can be any arbitrary topology, as long as the band is a simply-connected domain with faceted curves. The only exception where the band and moving objects can touch is a symmetry boundary.
- In **2D RZ**, the motion is constrained to the positive or negative Z-axis of the global coordinate system. The requirements are the same for motion in 2D XY, except the common Z-axis can have coincident band and moving objects.



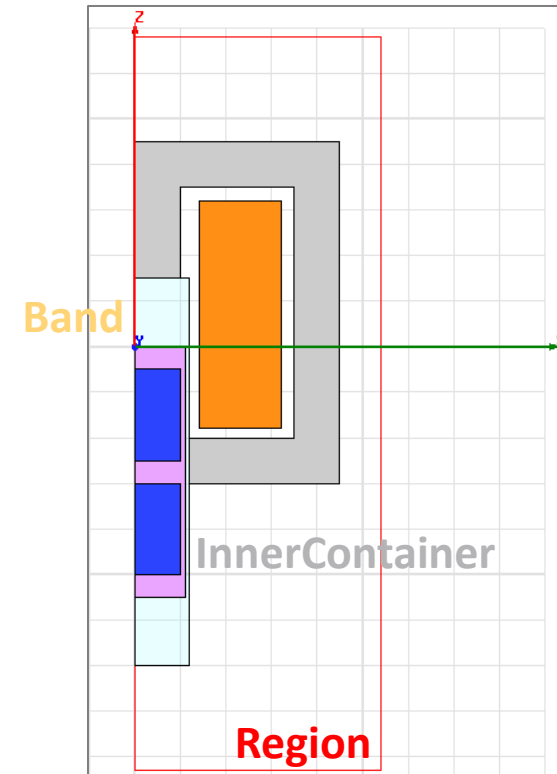
/ Translational Motion in 3D

- In 3D, the motion can be along the X-, Y-, or Z-axis for any defined coordinate system. The band and shape of the objects of motion can be any arbitrary topology, as long as the band is a simply-connected domain with faceted curves. The motion limits need to allow clearance for all moving objects for the entire extent of the motion limits.
- A common situation in 3D translational motion is a symmetry, which allows a half, quarter, sixth, etc. wedge of the geometry and simulation domain. The motion will be in a direction tangential to all symmetry planes. The band and moving objects will most likely be coincident along the symmetry planes. In that case it is important to remember to use the Symmetry multiplier in the Transient Design Settings.



/ Nomenclature

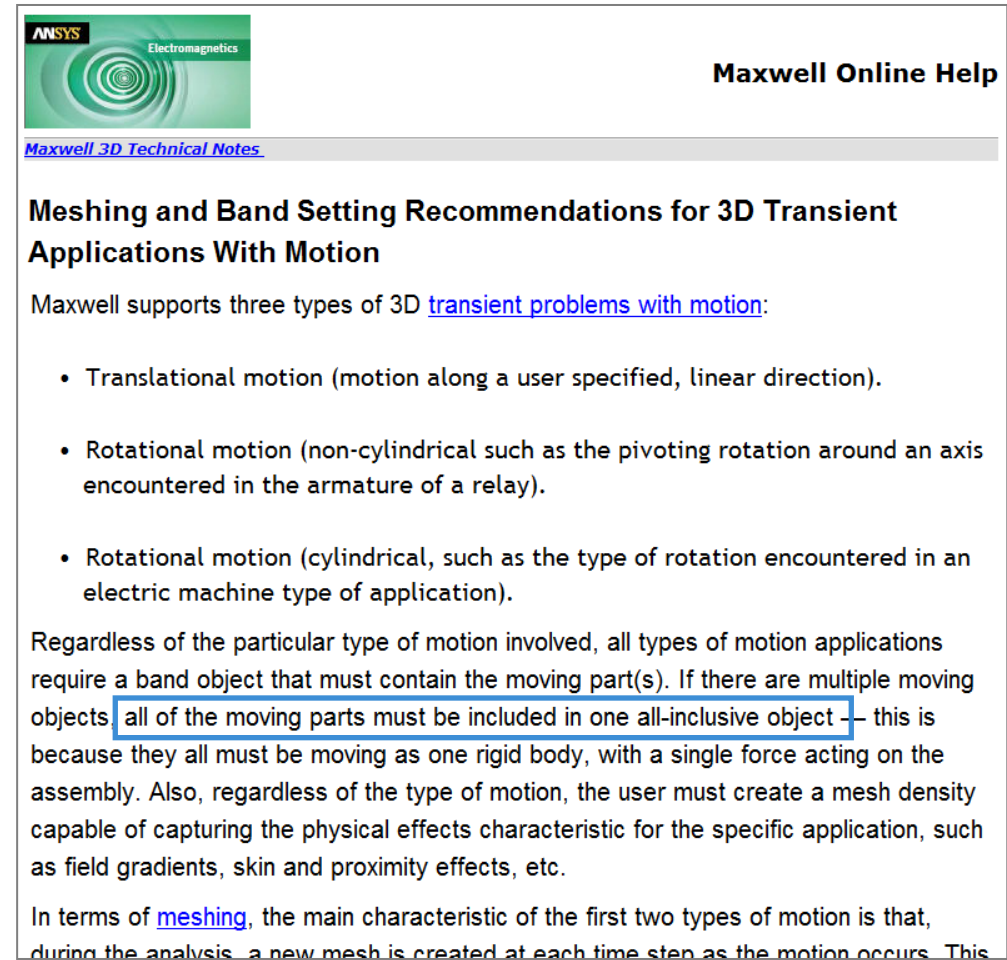
- **Band** – the object separating the moving from stationary objects
- **InnerContainer** – an object containing all the moving parts (within the Band)
- **OuterContainer** – an object outside of the band that might contain all objects (moving & stationary)
- **Region** – the object that defines the simulation domain



Note: These are all “vacuum” or “air” material objects. The names are all arbitrary, but commonly used. Their function is generally for meshing and solver purposes as explained in this document.

Some Recommendations can be found in the Help

- Please read the Online Help section for some preliminary recommendations in 3D
- The first recommendation is that the band *cannot be a cylindrical object*, but it must be segmented
- The second recommendation is to include multiple moving objects within “one all-inclusive object”, which can be called a *Container* or *InnerContainer* object. This is required for moving objects with holes or if there is more than one moving object
- Also, please note that the Band object must not touch stationary or moving objects, except for cases of symmetry



The screenshot displays the 'Maxwell Online Help' interface. At the top left is the ANSYS Electromagnetics logo. The title 'Maxwell Online Help' is at the top right. Below the logo is a link to 'Maxwell 3D Technical Notes'. The main heading is 'Meshing and Band Setting Recommendations for 3D Transient Applications With Motion'. The text states: 'Maxwell supports three types of 3D [transient problems with motion](#):' followed by a bulleted list: translational motion, non-cylindrical rotational motion, and cylindrical rotational motion. A paragraph explains that all motion applications require a band object containing the moving part(s), and that multiple moving objects must be included in one all-inclusive object. A final paragraph discusses meshing, noting that a new mesh is created at each time step as motion occurs.

ANSYS Electromagnetics

Maxwell Online Help

[Maxwell 3D Technical Notes](#)

Meshing and Band Setting Recommendations for 3D Transient Applications With Motion

Maxwell supports three types of 3D [transient problems with motion](#):

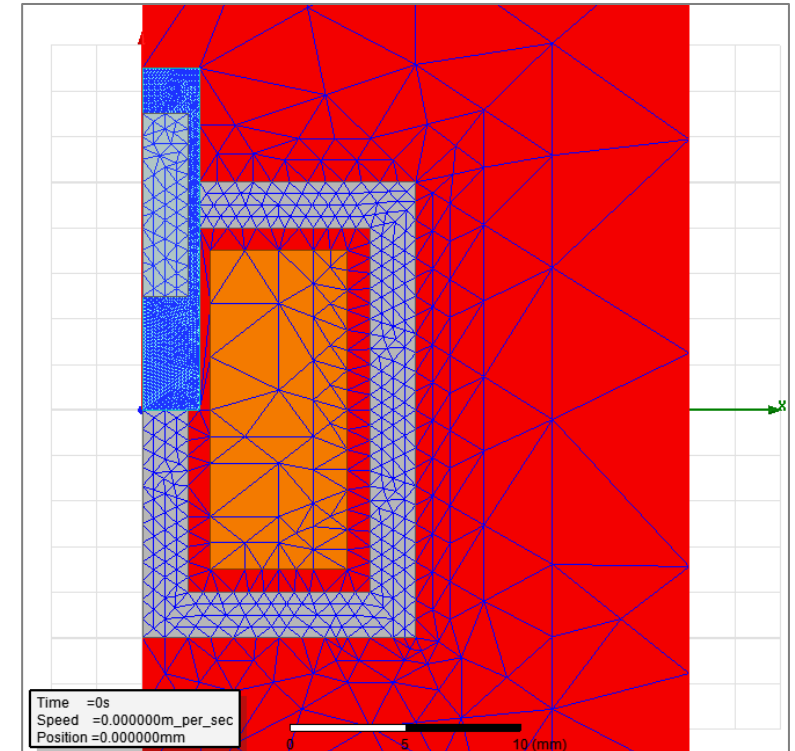
- Translational motion (motion along a user specified, linear direction).
- Rotational motion (non-cylindrical such as the pivoting rotation around an axis encountered in the armature of a relay).
- Rotational motion (cylindrical, such as the type of rotation encountered in an electric machine type of application).

Regardless of the particular type of motion involved, all types of motion applications require a band object that must contain the moving part(s). If there are multiple moving objects **all of the moving parts must be included in one all-inclusive object** — this is because they all must be moving as one rigid body, with a single force acting on the assembly. Also, regardless of the type of motion, the user must create a mesh density capable of capturing the physical effects characteristic for the specific application, such as field gradients, skin and proximity effects, etc.

In terms of [meshing](#), the main characteristic of the first two types of motion is that, during the analysis, a new mesh is created at each time step as the motion occurs. This

Reminders about Translational Motion

- The mesh between the **Band** and the **moving objects** is re-defined at every time-step with motion. The stationary mesh is constant for all time-steps, and the mesh inside the moving objects is also constant
- By using an **InnerContainer** object, we are also specifying the mesh constant inside the **InnerContainer** (it is a moving object), and the re-meshing occurs between the **InnerContainer** and the **Band**
- By using a simple **InnerContainer** object, the re-meshing is easier at each time-step
- Also, by controlling the mesh density on the **InnerContainer** and **Band**, it is possible to control the re-meshed density/quality for each time-step. This is important to get the desired accuracy in results



InnerContainer is a “Best Practice”

- The *InnerContainer* object can touch containing objects, but not the band (except for symmetry and/or the 2D Z-axis)
- In 2D, or if there is a large gap in 3D, you can space the container evenly with the Band within the air gap. This creates nice mesh layers
- With small gaps in 3D (especially near curved surfaces), place the faces of the container object on the outer faces of the objects, and space the band half-way in the air gap to avoid any additional complication of extra surfaces
- It is recommended that this InnerContainer be segmented, although it is not necessary
- The *InnerContainer* should not have any holes in it, but be as solid of an object as possible, that contains all the moving parts. It should also be as simple a shape as possible (a segmented cylinder) and fill as much of the moving space (with reasonable air gaps for meshing) so that the re-meshing procedure is as efficient as possible

OuterContainer is sometimes “Nice”

- The **OuterContainer** can touch any stationary objects outside of the band, but not the band itself (except for symmetry).
- The purpose of this object can be different – any of the following uses are good:
 - To “contain” the mesh by defining a volume in which to apply a loose mesh operation
 - To define a mesh operation within a specified volume. For this purpose, place the outer container around the volume of the desired objects (perhaps a sensor or a single winding).
 - To define a constant cylindrical surface outside of the band. This might be used if you have a cylindrical (segmented) band, but the stationary objects surrounding it have “teeth” that only sometimes are near the band. Using the **OuterContainer** in this way would keep a constant layer of mesh and may improve the mesh quality.

General Mesh Suggestions

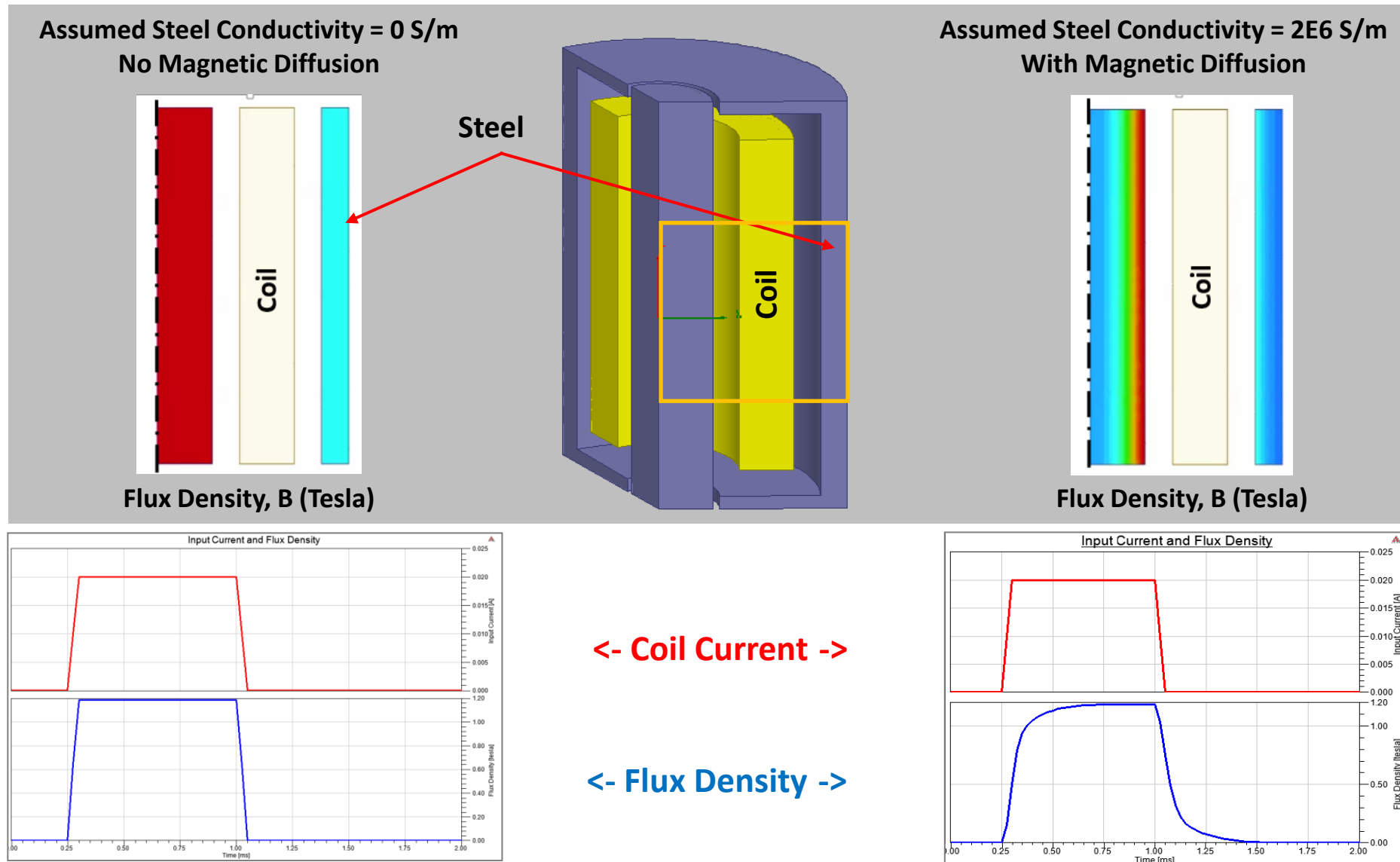
- *Length-based* Mesh operation *Inside* all magnetic objects (possibly including the coils) ~5-20k elements.
- *Length-based* Mesh operation *Inside* the Band and Container objects ~5-10k elements.
- Keep the Container object as simple as possible (segments, no holes, etc.)

Effects of Magnetic Field Diffusion

/ Effects of Eddy Currents: Magnetic Diffusion

- **Magnetic Diffusion:**
 - Permeable materials with non-zero conductivity will develop eddy currents when exposed to a time varying (step input) magnetic field. The induced eddy currents cause a delay in the build of magnetic flux, resulting in a time diffusion of the magnetic field through the thickness of the magnetic material.
- **Magnetic Diffusion Time:**
 - The time required for the fundamental component of flux density B at every point in a conducting material to rise from zero to at least approximately 63% of its final value after an applied step function $H(t)$.
 - Magnetic Infusion
 - Inward diffusion of magnetic flux upon turn-on of coil current
 - Magnetic Effusion
 - Outward diffusion of magnetic flux upon turn-off of coil current
- For ref. “**Magnetic Sensors and Actuators**” 2nd Edition, Chapter 5, J. Brauer, IEEE, Wiley Press

Effects of Eddy Currents: Magnetic Diffusion Illustration with FEA



/ Effects of Eddy Currents – Diffusion

- For stationary conducting linear materials, simple equations have been derived for magnetic diffusion times (delays of flux) in:
 - slabs of width w : $\tau_m = \mu \sigma (w/\pi)^2$
 - and cylinders of radius r : $\tau_m = \mu \sigma r^2 / (2.4048)^2$
- For saturated materials with a step B-H curve, simple equations have been derived:
 - slabs of width w : $\tau_m = \sigma w^2 B_m / (8H_o)$
 - and cylinders: $\tau_m = \sigma r^2 B_m / (4H_o)$
- For other shapes and B-H curves, finite element analysis is recommended. The Maxwell computations here also include effects of eddy currents caused by motion of the armature.
- See for reference: “[Magnetic Sensors and Actuators](#),” 2nd Edition, Chapter 5, J. Brauer, IEEE, Wiley Press

Advantages using the Transient Solver for Actuator Simulation

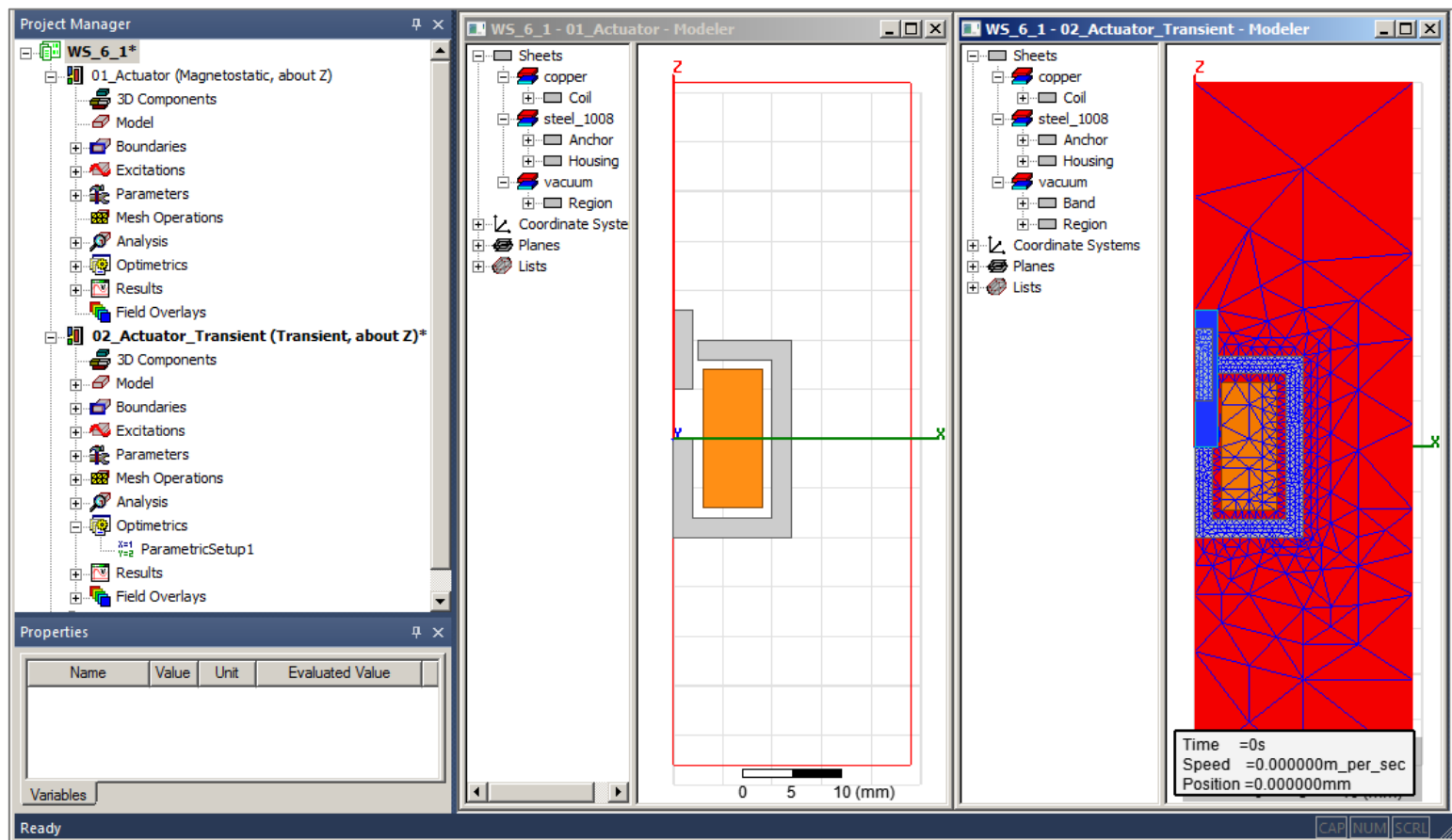
- Voltage sources, or coupled with an arbitrary circuit. The current is solved for based on magnetic inductance, diffusion, armature motion, switching circuits.
- Simultaneous electrical and mechanical time stepped solution allows users to define functional load and damping forces for springs, gravity, etc.
- Band object created to surround moving parts and used to conserve re-meshing for every time step.
- Eddy effects in conductive materials induced from coil current and armature motion.
- Magnetic diffusion
- Non linear material effects

Summary

What have we learned in this session?

- Actuators/solenoids modelling
- Translational motion settings
- Field diffusion effects
- Losses (See Appendix 6.2)

Workshop 06 – Actuator Example



Appendix 6.1 – Material Definitions



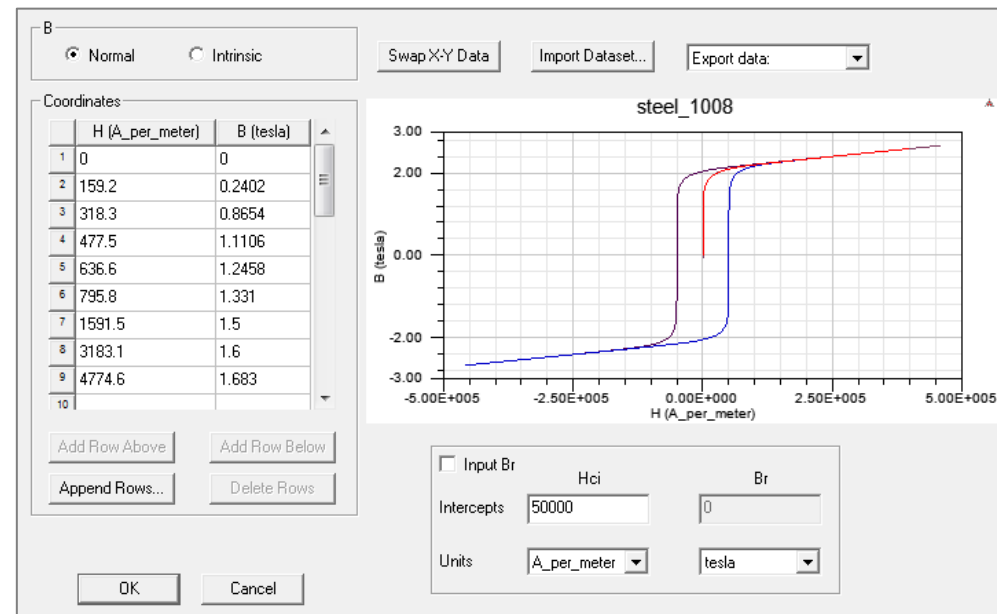
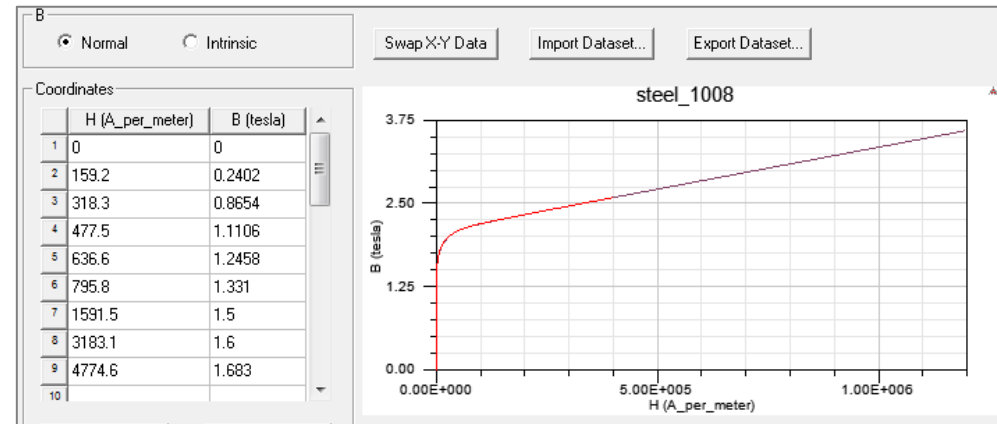
Material Definition

- **Transient Material Properties**
 - In a Magnetostatic simulation, the following parameters may be defined for a material
- **Relative Permeability :**
 - Relative permeability can be either Simple (linear μ_r) or Nonlinear (BH Curve) or/and anisotropic
The hysteresis loop can also be represented
- **Bulk Conductivity:**
 - Used to determine the current distribution in conductors
 - Can be Simple or Anisotropic
- **Magnetic Coercivity:**
 - Used to define permanent magnets properties.
 - Requires magnitude and direction specification.
 - Direction specified is with respect to Orientation CS of bodies to which material is assigned
- **Core Loss Type**
 - Can be Electrical Steel, Power Ferrite, Hysteresis Model or None
 - Allows to define Core Loss coefficients for loss calculation
- **Composition :**
 - Can be Solid or Lamination
 - Requires specification of direction normal to laminated pack and stacking factor

Material Definition

- Transient Material Properties

Properties of the Material				
	Name	Type	Value	Units
	Relative Permeability	Nonlinear	B-H Curve...	
	Bulk Conductivity	Simple	2000000	siemens/m
	Magnetic Coercivity	Vector		
	- Magnitude	Vector Mag	0	A_per_meter
	- X Component	Unit Vector	1	
	- Y Component	Unit Vector	0	
	- Z Component	Unit Vector	0	
	Core Loss Model		Electrical Steel	w/m^3
	- Kh	Simple	190	
	- Kc	Simple	2	
	- Ke	Simple	0.1	
	- Kdc	Simple	0	
	Mass Density	Simple	7872	kg/m^3
	Composition		Lamination	
	- Stacking Factor	Simple	0.95	
	- Stacking Direction		V(1)	



/ Material Definitions

- Material Properties

- Core Loss Type

- Core Loss Type can be either Electrical Steel or Power ferrite
 - Core Loss Coefficients will change according to selected Core Loss type

Core Loss Type		Electrical Steel	w/m^3
- Kh	Simple	0	
- Kc	Simple	0	
- Ke	Simple	0	
Mass Density	Simple	8055	kg/m^3

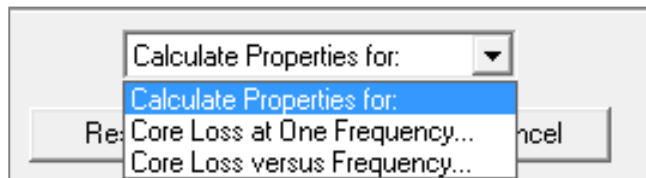
$$p_v = K_h f (B_m^2) + K_c (f B_m)^2 + K_e (f B_m)^{1.5}$$

Core Loss Type		Power Ferrite	w/m^3
- Cm	Simple	0	
- X	Simple	0	
- Y	Simple	0	
Mass Density	Simple	8055	kg/m^3

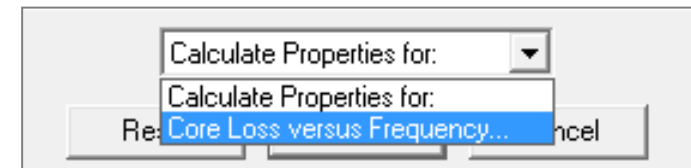
$$p_v = C_m f^x B_m^y$$

- Core Loss Coefficient Calculations

- Maxwell provides tools to evaluate core loss coefficients based on core loss data provided by users
 - Users can select tab at the bottom of View/Edit Material window and select the option “Calculate Properties for”



For Electrical Steel



For Power Ferrites

Material Definitions

- Material Properties

- Core Loss at One Frequency

- The option Calculate Properties for “Core Loss at One Frequency” enables to input B-P Curve for a defined frequency. This option is available only for Electrical Steel
 - Using specified B-P Curve, K_1 and K_2 coefficients are obtained by minimizing quadratic form

$$err(K_1, K_2) = \sum_i [P_{vi} - (K_1 B_{mi}^2 + K_2 B_{mi}^{1.5})]^2 = \min \quad \text{Where } i \text{ represents each point of defined B-P Curve}$$

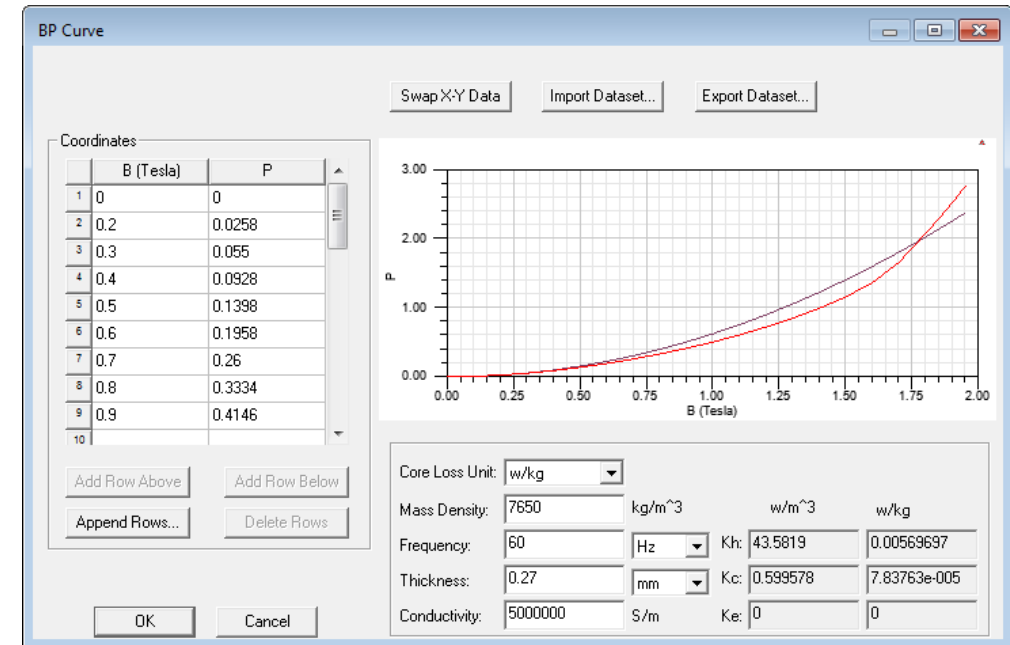
- Eddy Current coefficient, K_c is evaluated as

$$K_c = \pi^2 \sigma \frac{d^2}{\delta}$$

- Where, σ is the conductivity and d is the thickness of one lamination sheets and δ the skin depth.
 - Hysteresis Loss coefficient (K_h) and Excessive Loss Coefficient (K_e) are evaluated from K_1 , K_2 and

$$K_h = \frac{K_1 - K_c f_0^2}{f_0}$$

$$K_e = \frac{K_2}{f_0^{1.5}}$$



Material Definitions

- Material Properties

- Core Loss versus Frequency:

- Selecting the option Calculate Properties for “Core Loss versus Frequency” enables to input B-P Curve for multiple frequency values
- For Electrical Steel K_h , K_c and K_e are obtained by minimizing quadratic form

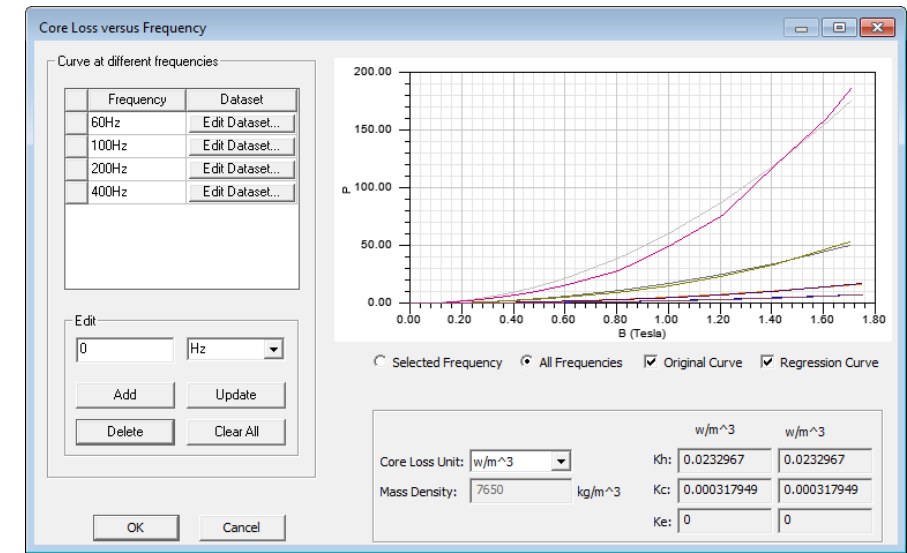
$$err(K_h, K_c, K_e) = \sum_{i=1}^m \sum_{j=1}^{n_i} [P_{vij} - (K_h f_i B_{mij}^2 + K_c f_i^2 B_{mij}^2 + K_e f_i^{1.5} B_{mij}^{1.5})]^2 = \min$$

Where, m is number of curves added and n_i is number of points defined in i^{th} curve

- For Power Ferrites C_m , x and y are obtained by minimizing quadratic form

$$err(c, x, y) = \sum_{i=1}^m \sum_{j=1}^{n_i} [\log(P_{vij}) - (c + x \log(f_i) + y \log(B_{mij}))]^2 = \min$$

$$c = \log(C_m)$$



Appendix 6.2 - Losses

/ Losses in Maxwell

- Types of losses
 - **Solid Loss**: joule losses due to currents flowing through conducting materials. Currents can be sources or induced eddy currents. Conducting materials are supposed to be “solid”
 - **Stranded Loss**: joule losses due to currents flowing through conducting materials. In this case conductors are set on Stranded and currents can be only sources
 - **Stranded LossR**: joule losses due to currents flowing through lumped resistance inserted in a Voltage-source winding
 - **Core Loss**
 - *Electrical Steel*: Steinmetz formula for laminated electrical steel
 - *Ferrite*: Steinmetz formula for ferrites
 - *Hysteresis model*: hysteresis losses calculated using actual hysteresis loop properties. Eddy Current losses calculated using Steinmetz formula.
 - **Total Loss**: losses averaged over two consecutive time steps

/ Solid Loss and Stranded Loss

- **Solid Loss** represents the resistive loss in 2D or 3D volume and it is calculated by:

$$SolidLoss = \iiint_{Vol} \frac{J^2}{\sigma} dV$$

Where σ is the material conductivity and J the conduction current density.

- **Stranded Loss** represents the resistive loss in 2D or 3D volumes when a conductor is set to **stranded** and the source is an external circuit:

$$StrandedLoss = \iiint_{Vol} \frac{J^2}{\sigma} dV$$

- The winding fill factor is assumed to be 100%.
- It is valid also for a current source placed directly in the winding settings.
- **StrandedLossR** represents the resistive loss based on I^2 times the lumped resistance R , when a conductor is set to **stranded** and the winding is set as voltage source:

$$StrandedLossR = RI^2$$

Core Loss

- **Core Loss** combines eddy current losses and hysteresis losses for a transient solution type
- This can be either a post-processing calculation, based on already calculated transient magnetic field quantities, or can also be integrated for every time step in an iterative algorithm, capable to evaluate the influence of such losses on the actual H-field and B-field distribution
- It is applicable to the evaluation of core losses in laminated steel or in power ferrites
- The Core loss model must be specified in Material properties
- The Core loss model is not available for 2D Eddy Current Analysis

Core Loss

- If in Eddy Current Solver the Steinmetz formulation is consistent, that is not anymore true for transient analysis, where motions and non linear materials are present
- Those two peculiarities make the concept of frequency to be inconsistent, and consequently also the validity of the formula drops
- In order to get around that issue, a modified version of Steinmetz formula is used in Transient, considering the B-Field time derivative

$$p_v = K_h f(B_m^2) + K_c (f B_m)^2 + K_e (f B_m)^{1.5} \rightarrow p_h(t) + p_c(t) + p_e(t) \quad \longleftrightarrow \quad f B_m = \frac{dB}{dt}$$

- where:

$$p_h(t) = \left\{ \left| H_x \frac{dB_x}{dt} \right|^{\frac{2}{\beta}} + \left| H_y \frac{dB_y}{dt} \right|^{\frac{2}{\beta}} + \left| H_z \frac{dB_z}{dt} \right|^{\frac{2}{\beta}} \right\}^{\frac{\beta}{2}}$$

$$p_e(t) = \frac{1}{C_e} K_e \left\{ \left(\frac{dB_x}{dt} \right)^2 + \left(\frac{dB_y}{dt} \right)^2 + \left(\frac{dB_z}{dt} \right)^2 \right\}^{0.75}$$

$$p_c(t) = \frac{1}{2\pi^2} K_c \left\{ \left(\frac{dB_x}{dt} \right)^2 + \left(\frac{dB_y}{dt} \right)^2 + \left(\frac{dB_z}{dt} \right)^2 \right\}$$

$$H_m = \frac{1}{\pi} K_h B_m$$

Total Loss

- **Total Loss** is associated with loss density fields in magnetic transient solutions
- Total loss uses the total integrated loss output from the solver and averages it over two consecutive time steps
- Total loss can be found as predefined quantity inside the Fields Calculator